



Cochin University of Science and Technology
Department of Computer Science

Seventh Semester M.Sc. (Five-Year Integrated) in Computer Science
(Artificial Intelligence & Data Science)

End Semester Examination - November 2024

21-805-0701 Computational Linguistics

Duration : 3 Hrs

Maximum Marks : 50

Answer all questions. From each question fully answer either (a) or (b)

1. (a) (i) Define Finite State Transducer (FST). Using an example, explain how an orthographic (e.g. e-insertion) rule in English can be implemented using FST. [6 marks][CO2, DL-2, BTL-3]
(ii) Discuss any two techniques used to handle unknown words and zero probabilities in statistical n-gram models. [4 marks][CO1, DL-2, BTL-3]
- OR
- (b) (i) Define Noisy Channel Model. Demonstrate with an example how this is used for non-word spelling correction. [6 marks][CO2, DL-2, BTL-3]
(ii) Discuss steps involved in morphological analysis of text. [4 marks][CO1, DL-2, BTL-2]
2. (a) Define Probabilistic Context Free Grammar (PCFG). Draw all possible parse trees for the sentence - *He saw the man with the telescope*, and identify the most probable one using the PCFG rules (with probabilities) given below. [10 marks][CO1, DL-1, BTL-2]

$S \rightarrow NP VP$ (4/4)	$V \rightarrow \text{saw}$ (3/4)		
$VP \rightarrow VP PP$ (1/5)	$V \rightarrow \text{is}$ (1/4)	$NP \rightarrow \text{she}$ (1/14)	$N \rightarrow \text{distance}$ (1/9)
$VP \rightarrow V NP$ (4/5)	$\text{Det} \rightarrow \text{the}$ (7/9)	$NP \rightarrow \text{here}$ (1/14)	$N \rightarrow \text{guitar}$ (1/9)
$NP \rightarrow \text{Det } N$ (9/14)	$\text{Det} \rightarrow \text{a}$ (2/9)	$PP \rightarrow P NP$ (3/3)	$N \rightarrow \text{telescope}$ (1/9)
$NP \rightarrow NP PP$ (2/14)	$N \rightarrow \text{man}$ (3/9)	$P \rightarrow \text{from}$ (1/3)	$N \rightarrow \text{flowers}$ (1/9)
$NP \rightarrow \text{he}$ (1/14)	$N \rightarrow \text{girl}$ (2/9)	$P \rightarrow \text{with}$ (2/3)	

OR

- (b) Explain CYK Algorithm for parsing using a suitable example.

[10 marks][CO1, DL-2, BTL-3]

- (a) (i) Explain Corpus Lesk algorithm for Word Sense Disambiguation with a suitable example. [5 marks][CO3, DL-2, BTL-2]
 (ii) Define semantics. Discuss any 4 desired characteristics for semantic representations of language text. [5 marks][CO3, DL-3, BTL-3]

OR

- (b) Define dense word embeddings. Using a suitable example, explain the steps involved in training a skip-gram Word2Vec model with negative sampling. [10 marks][CO3, DL-2, BTL-2]

4. (a) Define Pronoun Resolution. Explain Lappin and Leass algorithm for pronoun resolution using the example below. [10 marks][CO4, DL-2, BTL-2]

- (1) John saw a beautiful Acura Integra at the dealership.
 (2) He showed it to Bob.
 (3) He bought it

Use Saliency factors :- Sentence recency (100), Subject emphasis (80), Existential emphasis (70), Direct object emphasis (50), Indirect object and oblique complement emphasis (40) Non-adverbial emphasis (50), Head noun emphasis (80).

OR

- (b) (1) Define Text Coherence. Discuss the different types of coherent relations between two sentences defined in Hobb's algorithm. [5 marks][CO5, DL-2, BTL-3]
 (2) With suitable examples, discuss the four types of center-shifting transitions in discourse and their order of preference according to centering theory. [5 marks][CO5, DL-2, BTL-3]

5. (a) Define Statistical Machine Translation (SMT). Discuss IBM's statistical model for English-French translation using the example "And the program has been implemented" to "Le programme a ete mis en application" for the alignment {2, 3, 4, 5, 6, 6, 6} [10 marks][CO1, DL-2, BTL-2]

OR

- (b) Explain the three components of Phrase-based model for statistical machine translation. Briefly explain how they are used in decoding the correct translation. [10 marks][CO1, DL-1, BTL-2]



Cochin University of Science And Technology

Department of Computer Science

Seventh Semester Five Year Integrated MSc in Computer Science (Artificial Intelligence & Data Science)

End Semester Examination - November 2024

21-805-0707: Algorithmic Game Theory

Duration: 3 Hours

Maximum Marks: 50

Answer all questions. From each question fully answer either (a) or (b)

1. (a) What are the different strategies in game theory [5 marks][CO1, DL-2, BTL-2]

OR

(b) What do you mean by nash equilibrium? Explain the computational complexity of nash equilibrium

[5 marks][CO3, DL-4, BTL-2]

2. (a) Explain bayesian games with example

[5 marks][CO2, DL-3, BTL-4]

OR

(b) Explain subgame perfect equilibrium?

[5 marks][CO1, DL-3,

BTL-4]

3. (a) How can mechanisms be designed to ensure that agents provide truthful information about their arrival times, departure times, and valuations in dynamic online environments with discrete time periods, given the complexity introduced by uncertain events and strategic misreporting?

[5 marks][CO4, DL-4, BTL-4]

OR

(b) How does the structure of direct-revelation online mechanisms influence the effectiveness and efficiency of decision-making in dynamic environments with uncertain conditions?

[5 marks][CO3, DL-3, BTL-4]

4. (a) How do restrictions on early-arrival and late-departure misreports in online mechanisms impact the design of mechanisms to ensure incentive compatibility, and what are the practical implications of these restrictions for agents and mechanism designers?

[5 marks][CO3, DL-3, BTL-3]

OR

(b) Explain utilitarian and egalitarian objective function with Balls and Bins problem?

[5 marks][CO5, DL-3, BTL-4]

5. (a) Explain Epistemic game theory with example?
OR

[5 marks][CO3, DL-4, BTL-4]

(b) Explain the structure of MPC Protocol

[5 marks][CO4, DL-2, BTL-2]

6. (a) What are the key factors in a secure multiparty model?
OR

[5 marks][CO2, DL-3, BTL-4]

(b) How do MPC and Game Theory differ in their approaches to decision-making

[5 marks][CO2, DL-3, BTL-4]

7. (a) (i) Is VCG mechanism incentive compatible? Explain?

[3 marks][CO4, DL-3, BTL-4]

(ii) Define Clarke's pivot rule?

[2 marks][CO1, DL-1, BTL-1]

OR

(b) Explain cooperative game theory. What are the different models in cooperative games?

[5 marks][CO2, DL-2, BTL-2]

8. (a) Explain mechanism design without money?

[5 marks][CO3, DL-3, BTL-3]

OR

(b) Explain the house allocation problem?

[5 marks][CO2, DL-3, BTL-3]

9. (a) Explain voting? How can we evaluate the voting system?

[5 marks][CO1, DL-2, BTL-4]

OR

(b) Explain auction. What are the different types of auction?

[5 marks][CO3, DL-3, BTL-2]

10. (a) Explain strategic manipulation of votes?

[5 marks][CO2, DL-2, BTL-4]

OR

(b) Explain the winner's curse with an example?

[5 marks][CO5, DL-3, BTL-3]



Cochin University of Science And Technology
Department of Computer Science

Seventh Semester Master of Science (Five Year Integrated) in Computer Science
(Artificial Intelligence & Data Science)

End Semester Examination - November 2024

21-805-0703: Deep Learning

Duration: 3 Hours

Maximum Marks: 50

Answer all questions. From each question fully answer either (a) or (b)

1. (a)(i) Explain Multi-Layer Perceptron Algorithm [5 marks][CO1, DL-2, BTL-2]
(ii) What is Gradient based Learning? Explain and derive the idea behind Gradient Descent and Delta Rule. [5 marks][CO1, DL-2, BTL-3]
OR
(b)(i) Explain Back Propagation Algorithm along with the derivation considering both Cases of output and hidden unit. [7 marks][CO1, DL-2, BTL-3]
(ii) What is the role of Hidden Units in the networks? [3 marks][CO1, DL-1, BTL-2]
2. (a)(i) Explain the need of Regularization in deep learning. Illustrate different Regularization techniques [6 marks][CO2, DL-2, BTL-2]
(ii) What is meant by Data augmentation? Briefly explain types of Data augmentation. [4 marks][CO2, DL-1, BTL-2]
OR
(b) (i) Why do we need different learning rate for every features? [2 marks][CO2, DL-2, BTL-3]
(ii) Explain commonly used optimization techniques in detail. [8 marks][CO2, DL-2, BTL-2]
3. (a)(i) Explain the structure of the CNN architecture. What is the purpose of pooling in Convolutional Neural Network .What are the different types of pooling? [5 marks][CO3, DL-2, BTL-2]

(ii) Explain the working of Recurrent Neural Network. [5 marks][CO4, DL-2, BTL-2]
OR

(b) Explain the concept of LSTM and GRU with the help of the diagram [10 marks][CO4, DL-2, BTL-2]

4. (a) (i) Define the concept of Manifold Learning. Explain any 2 different Manifold Learning techniques [5 marks][CO5, DL-2, BTL-2]

(ii) Explain the concept of Probabilistic (PCA) and Independent component Analysis(ICA). [5 marks][CO5, DL-2, BTL-2]

OR

(b) (i) What is Multidimensional Scaling(MDS)? Explain different types of MDS. [6 marks][CO5, DL-2, BTL-3]

(ii) Define Autoencoders . Briefly explain different types of Auto encoders. [4 marks][CO5, DL-2, BTL-3]

5. (a) Explain the concept behind Boltzmann machines. Illustrate types of boltzmann machines [10 marks][CO6, DL-2, BTL-2]

OR

(b) (i) What is meant by Generative models?

[3 marks][CO6, DL-1, BTL-1]

(ii) Illustrate the working of Generative Adversarial Network (GAN).

[7 marks][CO6, DL-2, BTL-3]



COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

Programme: Master of Science (Five Year Integrated) in CS (AI&DS)
Course Code & Title 21-805-0707: Algorithmic Game Theory

Name of Examination: Series II		Max. Marks: 20	Semester: VII
Batch: 2021-26	Duration: 2 Hours	Date: 25.10.2024	Time: 10:00 AM – 12:00 PM

Answer all questions

1. What do you mean by cost sharing mechanism? (2 marks) [CO2, DL-1, BTL-1]
2. Explain rationality? (2 marks) [CO4, DL-2, BTL-1]
3. Explain epistemic game theory with example? (4 marks) [CO4, DL-2, BTL-2]
4. Explain Secure Multi-Party Computation(MPC) (2 marks) [CO4, DL-1, BTL-1]
5. Differentiate between cryptography and game theory (2 marks) [CO4, DL-2, BTL-2]
6. Explain the structure of MPC Protocol (2 marks) [CO3, DL-3, BTL-3]
7. Explain the common voting methods (2 marks) [CO3, DL-2, BTL-2]
8. Explain social welfare function and social choice function? (2 marks) [CO3, DL-2, BTL-2]
9. Explain Arrow's Theorem (2 marks) [CO3, DL-2, BTL-2]



Cochin University of Science and Technology

Department of Computer Science

Seventh Semester M.Sc. (Five-Year Integrated) in Computer Science (Artificial Intelligence & Data Science)

Second Series Examination - October 2024

Digital Image and Video Processing

Duration : 2 Hrs

Maximum Marks : 20

All questions are compulsory.

1. Explain the basic steps involved in frequency domain filtering along with filter transfer function for each of the low pass filter

[4 marks][CO3, DL-1, BTL-2]

2. Compare between line detection and edge detection. Give different line detection masks.

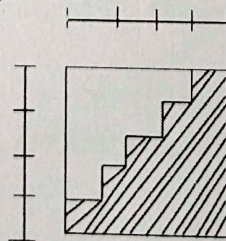
[3 marks][CO4, DL-1, BTL-2]

3. Consider an image that uses a window of size 5 X 5. The gray level values inside the 5 X 5 Sub-image are 15,17,15,17,16,10,8,9,18,15,16,12,14,11,15,14,15,11,100,15,14,13,12,12,17. What values could (a) A local averaging filter (mean) (b) A median filter (c) A mode filter (d) A max filter (e) A min filter assign to the central pixel of this sub-image

[3 marks][CO3, DL-2, BTL-3]

4. Give the methodology and algorithm for split and merge. Apply split and merge technique to segment the image given in Figure 4 (b), also represent quadtree representation of the segments.

[5 marks][CO5, DL-2, BTL-3]



5. What is Edge Detection? How gradient can be used for edge detection. Derive Roberts Cross Gradient operator.

[3 marks][CO4, DL-2, BTL-3]

6. Write note on homomorphic filtering

[2 marks][CO4, DL-2, BTL-2]



Cochin University of Science And Technology

Department of Computer Science

Seventh Semester Five Year Integrated M.Sc in Computer Science (Artificial Intelligence and Data Science)

Second Series Examination - October 2024

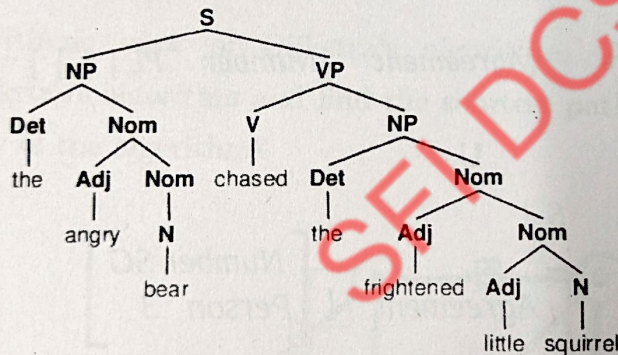
21-805-0701 Computational Linguistics

Duration: 2 Hours

Maximum Marks: 20

Answer all questions.

1. Define Context Free Grammar (CFG). Write the CFG used to generate the given tree. [4 marks][CO2, DL-1, BTL-2]



2. Explain Syntactic Ambiguity in Parsing. List any 3 types of syntactic ambiguity in English with suitable examples. Mention what the ambiguity is in each example and why it belongs to a particular type. [4 marks][CO2, DL-2, BTL-3]
3. Define Term-Frequency - Inverse Document Frequency (TF-IDF) vector for word representation. Compute the TF-IDF vector for the words {cat, rat, bird} using the corpus below. Using the cosine similarity, find which one of the two words is most similar to 'cat'.

Document 1: "The cat chased the rat", "The rat hid inside the tree"

Document 2: "So cat went looking for the bird instead"

Document 3: "Now cat, rat, and bird are thick friends"

[4 marks][CO1, DL-2, BTL-3]



COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

Programme: Master of Science (Five Year Integrated) in CS (AI&DS)
Course Code & Title 21-805-0707: Algorithmic Game Theory

Name of Examination: Series II		Max. Marks: 20	Semester: VII
Batch: 2021-26	Duration: 2 Hours	Date: 25.10.2024	Time: 10:00 AM – 12:00 PM

Answer all questions

1. What do you mean by cost sharing mechanism? (2 marks) [CO2, DL-1, BTL-1]
2. Explain rationality? (2 marks) [CO4, DL-2, BTL-1]
3. Explain epistemic game theory with example? (4 marks) [CO4, DL-2, BTL-2]
4. Explain Secure Multi-Party Computation(MPC) (2 marks) [CO4, DL-1, BTL-1]
5. Differentiate between cryptography and game theory (2 marks) [CO4, DL-2, BTL-2]
6. Explain the structure of MPC Protocol (2 marks) [CO3, DL-3, BTL-3]
7. Explain the common voting methods (2 marks) [CO3, DL-2, BTL-2]
8. Explain social welfare function and social choice function? (2 marks) [CO3, DL-2, BTL-2]
9. Explain Arrow's Theorem (2 marks) [CO3, DL-2, BTL-2]



COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

Programme : Seventh Semester MSc (Five Year Integrated) in Computer Science Artificial Intelligence & Data Science
Course Code & Title: 21-805-0703 Deep Learning

Name of Examination: Series I	Max. Marks: 20	Semester: VII
Batch: 2023-28	Duration: 2 Hours	Date: 10.09.2024
		Time: 10:00 AM – 12:00 PM

Answer all questions

- Describe the Backpropagation algorithm (3 marks) [CO1, DL-1, BTL-2]
 - Derive the equation for updating weights in back propagation algorithm (3 marks)[CO1, DL-2, BTL-3]
- Why do we need a different learning rate for every feature? Which are the different optimization techniques used to adapt the different learning rate for each parameters? (3.5marks)[CO2, DL-3, BTL-3]
- What is meant by regularization in machine learning? Briefly explain the commonly used regularization techniques. (3 marks)[CO2, DL-2, BTL-2]
- Describe Multi-Layer Perceptron Learning Algorithm. (3 marks)[CO1,DL-2,BTL-2]
- Explain the structure of the CNN architecture. What is the purpose of pooling in a CNN? What are the different types of pooling? (3.5 marks)[CO3, DL-2,BTL-3]
 - An input image has been converted into a matrix of size 12×12 along with the filter size 3×3 with a stride of 1. Then determine the size of the convoluted matrix. (1 mark)[CO3,DL-1,BTL-3]



COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

Programme: Seventh Semester Master of Science (Five Year Integrated) in Computer Science (Artificial Intelligence & Data Science)
Course Code & Title: 21-805-0703 Deep Learning

Name of Examination: Series II		Max. Marks: 20	Semester: VII
Batch: 2023-28	Duration: 2 Hours	Date: 24.10.2024	Time: 10:00 AM – 12:00 PM

Answer all questions

1. What is meant by Manifold Learning? Explain any two Manifold Learning Techniques.
(5 marks) [CO6, DL-2, BTL-2]
2. Briefly Explain Independent Component Analysis
(2 marks) [CO6, DL-1, BTL-1]
3. What is Auto Encoders? Describe different types of Auto Encoders.
(5 marks) [CO6, DL-2, BTL-2]
4. Describe the concept of LSTM in detail.
(4 marks) [CO5, DL-2, BTL-2]
5. Explain Gated Recurrent Unit.
(4 marks) [CO5, DL-2, BTL-3]



COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

Programme: Master of Science (Five Year Integrated) in Computer Science (AI and DS)
Course Code & Title: 23-813-0712: Complex Network Analysis

Name of Examination: Series II		Max. Marks: 20	Semester: VII
Batch: 2021-26	Duration: 2 Hours	Date: 24-10-2024	Time: 2:00 PM – 4:00 PM

Answer all questions

1. What are the key components and steps involved in a social influence analysis algorithm and explain how does it quantify the impact of individuals within a social network.
(6 marks) [CO7, DL-2, BTL-3]
2. Explain the different approaches for link prediction in network analysis
(4 marks) [CO8, DL-1, BTL-2]
3. What are the fundamental steps in applying spectral partitioning to divide a network and how does the eigen vector selection influence the quality of the partitioning.
(3 marks) [CO6, DL-1, BTL-2]
4. For a given graph $G(V,E)$, where $V = \{1,2,3,4,5,6\}$ $E = \{(1,2),(1,3),(2,3),(4,5),(5,6)\}$. The graph has two distinct communities $C1 = \{1,2,3\}$ and $C2 = \{4,5,6\}$. Calculate the modularity of this graph with respect to the given community.
[3 marks] [CO7, DL-1, BTL-2]
5. What are the factors that influence the flow of information in social network depends on
[4marks] [CO8, DL-1, BTL-2]



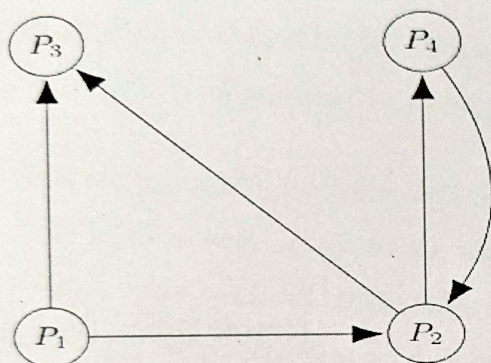
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

Programme: M.Sc. (Five Year Integrated) in Computer Science
Course Code & Title: 23-813-0712: Complex Network Analysis

Name of Examination: Series I	Max. Marks: 20	Semester: VII
Batch: 2021-26	Duration: 2 Hours	Date: 10-09-2024
		Time: 2:00 PM – 4:00 PM

Answer all questions

1. Using the following web graph find the steady-state PageRank vector. (Assume that $\alpha = .85$)

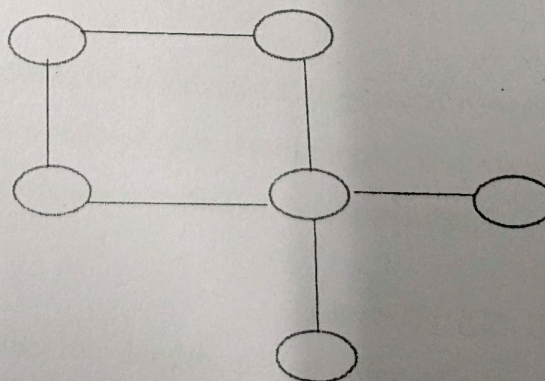


(4 marks) [CO1, DL-2, BTL-3]

2. Explain the Social network theory that focuses on the analysis of relationships, connections, and patterns of interaction among social groups.

(2 marks) [CO1, DL-1, BTL-2]

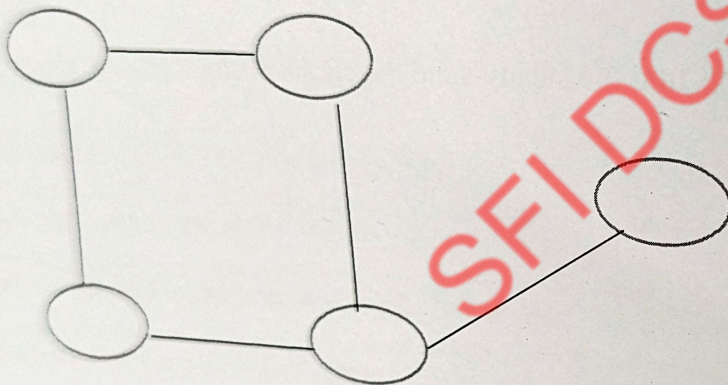
3. Calculate the assortativity index of the graph given below. Determine whether the graph is assortative, disassortative or neither.



(3 marks) [CO1, DL-1, BTL-2]

4. Explain the characteristics of Subcritical regime and Supercritical regime of random graph.
[2 marks] [CO2, DL-1, BTL-2]
5. Describe small world property. Illustrate the generation of a small word model from a regular lattice.
[3marks] [CO2, DL-1, BTL-2]
6. Consider a random network generated according to $G(n, p)$ model, where the total number of nodes is 50 and the probability that there are links between any two nodes is .60. Determine the probability that there are exactly 25 links in the network.
[3 marks] [CO2, DL-2, BTL-2]

7. Calculate the closeness centrality of the given graph.



[3 marks] [CO1, DL-2, BTL-2]



COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

Programme: Seventh Semester Five Year Integrated MSc in Computer Science (Artificial Intelligence & Data Science)

Course Code & Title 21-805-0707: Algorithmic Game Theory

Name of Examination: Series I	Max. Marks: 20	Semester: VII
Batch: 2021-26	Duration: 2 Hours	Date: 11.09.2024
		Time: 10:00 AM – 12:00 PM

Answer all questions

1. In the context of the Prisoner's Dilemma and the Pollution Game, how do the dominant strategies affect the overall outcome for all players involved? (2 marks) [CO4, DL-1, BTL-4]
2. Suppose two sellers are selling a product to three buyers, where Buyers A and C can only purchase from Sellers 1 and 2, respectively, while Buyer B can buy from either seller depending on which offers a lower price. All buyers have a budget of 1, meaning they will not purchase if the price exceeds 1. The sellers must set their prices within the range $[0, 1]$, and Buyer B will choose the cheaper seller (with ties being broken in favour of Seller 1). What type of equilibrium, if any, exists in this pricing game between the two sellers? Does the game have a pure strategy Nash equilibrium, a mixed strategy Nash equilibrium, or neither? (2 marks) [CO2, DL-2, BTL-2]
3. What is the concept of a correlated equilibrium, and how does it differ from a Nash equilibrium? Explain with an example (2 marks) [CO1, DL-1, BTL-1]
4. Define One-Shot Simultaneous Move Games. (2 marks) [CO1, DL-1, BTL-1]
5. What does it mean for an online mechanism to be dominant-strategy incentive-compatible (DSIC) given limited misreports, and how does this concept differ from Bayes-Nash incentive compatibility (BNIC) in terms of agent strategy and the stochastic nature of the mechanism? (2 marks) [CO2, DL-2, BTL-2]
6. Define single-valued online domain and what role does the partial order $\geq$ play in the design and analysis of truthful mechanisms in this context? (2 marks) [CO2, DL-1, BTL-2]
7. What is Pigou's example in the context of selfish routing, and how does the price of anarchy and the price of stability compare in this example? (2 marks) [CO1, DL-2, BTL-1]

8. Consider a game with m players and m machines. Each player chooses a machine, and the objective is to minimize the total cost, where the cost function is typically the maximum load across all machines. Prove that the price of anarchy of this game with respect to the utilitarian objective function is precisely $2 - 1/m$. (2 marks) [CO3, DL-3, BTL-3]
9. How does the structure of direct-revelation online mechanisms, including the definition of the mechanism state h_t , the handling of stochastic events, and the specification of decision and payment policies, influence the effectiveness and efficiency of decision-making in dynamic environments with uncertain conditions? (2 marks) [CO1, DL-2, BTL-1]
10. How do restrictions on early-arrival and late-departure misreports in online mechanisms impact the design of mechanisms to ensure incentive compatibility, and what are the practical implications of these restrictions for agents and mechanism designers? (2 marks) [CO1, DL-2, BTL-1]



COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

Programme: Seventh Semester Five Year Integrated MSc in Computer Science (Artificial Intelligence & Data Science)

Course Code & Title 21-805-0707: Algorithmic Game Theory

Name of Examination: Series II	Max. Marks: 20	Semester: VII
Batch: 2021-26	Duration: 2 Hours	Date: 25.10.2024
		Time: 10:00 AM – 12:00 PM

Answer all questions

1. What do you mean by cost sharing mechanism? (2 marks) [CO2, DL-1, BTL-1]
2. Explain rationality? (2 marks) [CO4, DL-2, BTL-1]
3. Explain epistemic game theory with example? (4 marks) [CO4, DL-2, BTL-2]
4. Explain Secure Multi-Party Computation(MPC) (2 marks) [CO4, DL-1, BTL-1]
5. Differentiate between cryptography and game theory (2 marks) [CO4, DL-2, BTL-2]
6. Explain the structure of MPC Protocol (2 marks) [CO3, DL-3, BTL-3]
7. Explain the common voting methods (2 marks) [CO3, DL-2, BTL-2]
8. Explain social welfare function and social choice function? (2 marks) [CO3, DL-2, BTL-2]
9. Explain Arrow's Theorem (2 marks) [CO3, DL-2, BTL-2]



Cochin University of Science and Technology
Department of Computer Science

**Seventh Semester M.Sc. (Five-Year Integrated) in Computer Science
(Artificial Intelligence & Data Science)**

First Series Examination - September 2024

21-805-0702: Digital Image and Video Processing

Duration : 2 Hrs

Maximum Marks : 20

All questions are compulsory.

1. Define a digital image.
[1 marks][CO1, DL-1, BTL-2]
2. Define a) Impulse Sequence b) Periodic Sequence
[2 marks][CO1, DL-1, BTL-2]
3. Give the formulation and plot of a) Contrast stretching b) Thresholding c) Image negative d) Log transformation e) Power law transformation Neatly label the x and y axis. or explain basic gray level transformation function
[3 marks][CO2, DL-1, BTL-2]
4. Using second derivative develop Laplacian mask for image sharpening
[3 marks][CO2, DL-2, BTL-2]
5. Compare between unsharp masking and high boost filtering.
[2 marks][CO2, DL-1, BTL-2]
6. Given the following kernel and image perform convolution of the two? $w = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ $f = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$
[3 marks][CO2, DL-2, BTL-3]
7. Find all bitplanes of the following 4 bit image $f = \begin{bmatrix} 0 & 3 & 7 & 8 \\ 1 & 1 & 2 & 2 \\ 3 & 14 & 15 & 13 \\ 4 & 6 & 9 & 11 \end{bmatrix}$
[2 marks][CO2, DL-2, BTL-3]
8. Find 2D DFT of the following image $f(x, y) = \begin{bmatrix} 2 & 4 \\ 7 & 3 \end{bmatrix}$
[4 marks][CO1 DL-3, BTL-3]



COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

Programme: Seventh Semester Five Year Integrated MSc in Computer Science (Artificial Intelligence & Data Science)

Course Code & Title: 21-805-0703 Deep Learning

Name of Examination: Series I		Max. Marks: 20	Semester: VII
Batch: 2023-28	Duration: 2 Hours	Date: 10.09.2024	Time: 10:00 AM – 12:00 PM

Answer all questions

- (i) Describe the Backpropagation algorithm (3 marks) [CO1, DL-1, BTL-2]
 - (ii) Derive the equation for updating weights in back propagation algorithm (3 marks)[CO1, DL-2, BTL-3]
2. Why do we need a different learning rate for every feature? Which are the different optimization techniques used to adapt the different learning rate for each parameters? (3.5marks)[CO2, DL-3, BTL-3]
3. What is meant by regularization in machine learning? Briefly explain the commonly used regularization techniques. (3 marks)[CO2, DL-2, BTL-2]
4. Describe Multi-Layer Perceptron Learning Algorithm. (3 marks)[CO1,DL-2,BTL-2]
5.
 - (i) Explain the structure of the CNN architecture. What is the purpose of pooling in a CNN? What are the different types of pooling? (3.5 marks)[CO3, DL-2,BTL-3]
 - (ii) An input image has been converted into a matrix of size 12x12 along with the filter size 3x3 with a stride of 1. Then determine the size of the convoluted matrix. (1 mark)[CO3,DL-1,BTL-3]